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ORIGINAL RESEARCH

The Efficacy and Safety of Liraglutide in Patients Remaining Obese 6 Months after Metabolic Surgery

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ABSTRACT

Introduction: The safety and efficacy of liraglutide as a weight loss intervention in individuals who remain obese within 1 year post-metabolic surgery remain unclear. This study aimed to evaluate the effects and safety of liraglutide (1.8 mg) in patients with persistent obesity at 6 months postoperatively.

Methods: This retrospective cohort study included 61 patients who remained obese (body mass index [BMI] ≥ 28.0 kg/m^2) at 6 months postoperatively. Among these patients, 27 were treated with 1.8 mg of liraglutide for 12 weeks, whereas 34 served as controls. The primary endpoint was the change in total weight loss (%TWL) after 24 weeks. Changes in weight, BMI, complications, and adverse events were also assessed.

Results: The liraglutide group showed a greater reduction in %TWL than the control group (11.6% ± 1.1% vs. 4.9% ± 1.0%), with an estimated treatment difference of 6.6% (95% confidence interval [CI], 3.7–9.6%, $P < 0.01$). The adjusted mean differences in the reduction

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of weight and BMI between the liraglutide and control groups were – 6.2 kg (95% CI – 8.9 to – 3.4, $P<0.01$) and – 3.0 kg/m^2 (95% CI – 4.2 to – 1.7, $P<0.01$), respectively. The liraglutide group exhibited increased rates of remission in non-alcoholic fatty liver disease and hypertension. No serious adverse reactions were observed. ***Conclusions***: For patients who remained obese at 6 months postoperatively, 12-week liraglutide treatment resulted in increased weight loss, improved metabolic control, and high rate of remission for obesity-related metabolic diseases after 24 weeks. Earlier and more timely adjuvant weight loss medication intervention based on BMI within 1 year postoperatively may enhance weight loss after metabolic surgery.
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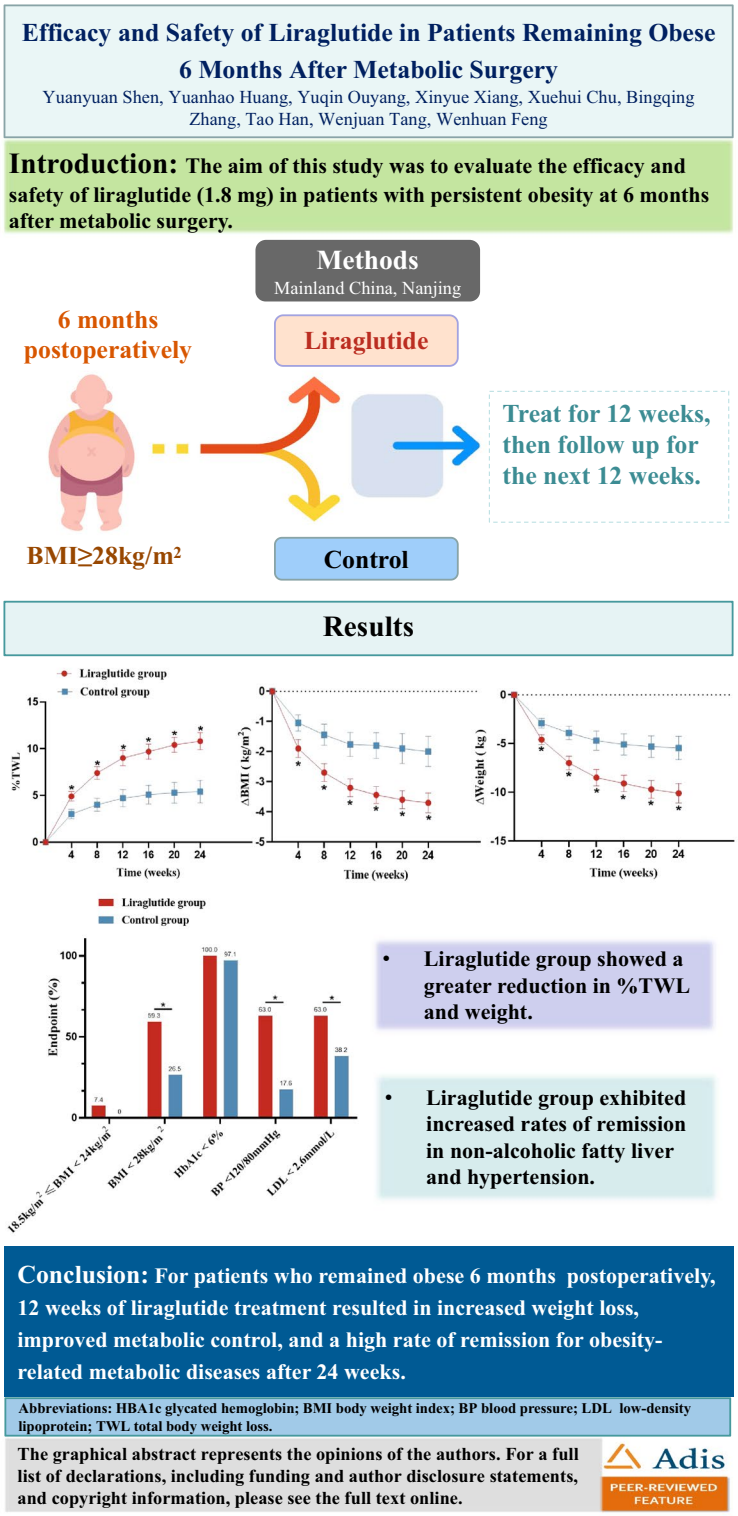
Graphical Abstract:



Keywords: Body weight; Liraglutide; Metabolic diseases; Metabolic surgery; Obesity

DIGITAL FEATURES

This article is published with digital features, including graphical abstract, to facilitate understanding of the article. To view digital features for this article, go to https://doi.org/10.6084/m9.figshare.26729122.

Key Summary Points
Why carry out this study?
Approximately 25% of patients experience inadequate weight loss 2 years after metabolic surgery, and subsequent weight loss interventions for those with inadequate weight loss > 1 year postoperatively result in limited weight loss
Patients who remain obese after metabolic surgery are more likely to experience inadequate weight loss, and > 80% weight loss within the first year after surgery is achieved within the first 6 months
Adjuvant weight loss medication based on body mass index (BMI) within 1 year postoperatively may potentially optimize postoperative weight loss; however, studies in this area remain lacking
What was learned from the study?
Liraglutide treatment for 12 weeks resulted in increased weight loss and a high remission of obesity-related metabolic diseases
This study suggests a more timely adjuvant weight loss medication intervention based on BMI after metabolic surgery

INTRODUCTION

Metabolic surgery is the most effective treatment for losing weight and alleviating the metabolic disease burden in patients with moderate-to-severe obesity, particularly when medical therapies fail [1]. One year after metabolic surgery, the lowest body weight period, > 90% of patients attain a successful weight loss goal defined as total weight loss (%TWL) > 20%; nonetheless, approximately 30% of patients remain obese, with a body mass index (BMI) of ≥ 28 kg/m^2, the defined threshold for obesity in China [2]. Subsequently, with the cessation of weight loss and regain, inadequate weight loss identified as %TWL ≤ 20% affects up to 25% of patients after 2 years [3, 4].

Maximizing weight loss after metabolic surgery is pivotal for patients with moderate-to-severe obesity. Individuals with greater weight loss and lower postoperative BMI have a lower possibility of weight regain and achieve greater and longer remission rates for obesity-related diseases [5, 6]. Compared with other patients, patients who remain obese 1 year after metabolic surgery are more likely to regain weight because of relapse to preoperative poor eating habits and sedentary lifestyle. These patients also have inadequate weight loss thereafter [7, 8]. Current additional weight loss interventions focus on behavioral approaches, pharmacotherapy, endoscopic procedures, and modified metabolic surgery for patients experiencing inadequate weight loss or weight regain, who often have a high BMI close to their preoperative BMI and therefore have limited weight loss [9–14]. Limited data exist on the efficacy of pharmacotherapy in treating patients who remain obese postoperatively [15].

Previous studies reported that > 80% of weight loss in the first postoperative year was achieved within the first 6 months, whereas < 20% of weight loss was observed in the following 6 months because of a gradual increase in caloric intake [16–18]. Our previous study showed that 42.9% of individuals with a BMI of 24–28 kg/m^2 at 6 months achieved healthy weight (18.5 kg/m^2 ≤ BMI < 24 kg/m^2) at 12 months postoperatively. However, only 3.4% of patients with obesity at 6 months attained this target (Supplementary Material, Fig. S1). However, no current trial has yet to demonstrate the efficacy and safety of incorporating liraglutide intervention in individuals who remain obese 6 months after metabolic

surgery. Thus, earlier additional weight loss interventions might be considered for patients with obesity at 6 months postoperatively prior to weight loss plateau at 1 year, considering that the likelihood of achieving significant weight loss decreases subsequently.

Liraglutide, a glucagon-like peptide-1 (GLP-1) receptor agonist, is used in the treatment of obesity by primarily inducing weight loss via appetite inhibition [19]. Previous reports indicated that liraglutide led to notable weight loss and improved metabolic profiles in patients who experienced inadequate weight loss or weight regain > 1 year after metabolic surgery [9, 20, 21]. However, no current trial has yet to demonstrate the efficacy and safety of incorporating liraglutide intervention in individuals who remain obese 6 months after metabolic surgery.

Therefore, the present study aimed to evaluate whether the addition of liraglutide intervention would lead to greater weight loss, improvements in metabolism, and mitigation of metabolic diseases, as compared with solely implementing dietary and exercise interventions for patients experiencing obesity 6 months postoperatively.

METHODS

Study Design

This retrospective cohort study was conducted at the Affiliated Drum Tower Hospital, Medical School, Nanjing University, China, from November 2021 to March 2023. Patients who remained obese 6 months after metabolic surgery and finished 1 year of follow-up were enrolled. The inclusion criteria were as follows: individuals aged ≥ 18 years, underwent primary metabolic surgery ≥ 1 year, had a BMI of ≥ 28 kg/m^2 at 6 months postoperatively, and could provide informed consent. The exclusion criteria were as follows: presence of an anatomical or endocrinological pathology causing suboptimal weight loss or weight regain (e.g., gastrogastric fistula), calcitonin concentrations of ≥ 50 ng/l, a family or personal history of multiple endocrine neoplasia type 2 or familial medullary thyroid carcinoma, a history of acute or chronic pancreatitis, an estimated glomerular filtration rate of < 30 ml/min/1.73 m^2, current pregnancy, inability to maintain adequate contraception, breastfeeding, a personal history of heart failure with New York Association class III or higher, or a personal history of diabetic gastroparesis. The trial was conducted in accordance with the principles embodied in the Declaration of Helsinki and was approved by the Ethics Committee of the Affiliated Drum Tower Hospital, Medical School, Nanjing University (approval no. 2017-030-08). All participants provided informed consent for their data potentially being used in the study, and the anonymity of all participants was preserved. Similar methods have been presented [6].

Study Population

Patients eligible for the trial were divided into two groups based on their treatment regimen at 6 months postoperatively: (1) control group, which underwent the diet–exercise program, and (2) liraglutide group, which underwent the diet-exercise program plus liraglutide therapy at 6 months postoperatively. In routine clinical practice in our center, patients with a BMI of ≥ 28 kg/m^2 at 6 months postoperatively were given an extra diet–exercise program as follows and were advised to receive liraglutide treatment. The starting dose of liraglutide was 0.6 mg per day administered subcutaneously at week 1, 1.2 mg per day as tolerated at week 2, and up to 1.8 mg per day from week 3 to the end of week 12.

The exercise program was designed to meet the World Health Organization (WHO) recommendations of a minimum of 75 min/week of vigorous-intensity aerobic physical activity, 150 min/week of moderate-intensity aerobic physical activity, or an equivalent combination of both [22]. Participants followed a low-calorie diet of 1200 kcal/day. Patients were advised to take one Centrum®, Caltrate®, and alfacalcidol soft capsule (0.5 μg) twice daily. Patients received monthly telephone follow-ups, and

follow-up appointments were scheduled in the clinic at 12 and 24 weeks.

Participants with type 2 diabetes mellitus (T2DM) were diagnosed according to the WHO criteria, and the remission of T2DM was defined as having glycosylated hemoglobin (HbA1c) levels of < 6%, fasting blood glucose (FBG) levels of < 5.55 mmol/l, and the absence of anti-diabetic medications [23, 24]. Hypertension was defined as systolic blood pressure (SBP) of ≥ 140 mmHg, diastolic blood pressure of ≥ 90 mmHg, or the use of antihypertensive medication. Additionally, the remission of hypertension was defined as blood pressure (BP) of < 120/80 mmHg without antihypertensive medication [25]. Non-alcoholic fatty liver disease (NAFLD) was diagnosed using abdominal ultrasound, the degree of lipid deposition and liver fibrosis was evaluated using the controlled attenuation parameter, and liver stiffness was measured using FibroTouch® (Wuxi Heskell Medical Technology Co., Ltd., Wuxi, China) [26]. Hyperuricemia was diagnosed with serum uric acid (UA) levels of ≥ 420 mmol/l or using anti-hyperuricemia medications [27]. Furthermore, dyslipidemia was defined as fasting total lipoprotein cholesterol levels of ≥ 5.2 mmol/l, low-density lipoprotein cholesterol (LDL-C) levels of ≥ 3.4 mmol/l, high-density lipoprotein cholesterol (HDL-C) levels of < 1.04 mmol/l, triglyceride levels of ≥ 1.7 mmol/l, or a history of lipid-lowering agents [25]. The remissions of dyslipidemia and hyperuricemia were defined as achieving normal biochemical values for serum lipid and UA levels without medication.

Data Collection

Participants' data, including height, weight, BP, comorbid conditions, and medication usage, were collected at baseline and at 4, 8, 12, 16, 20, and 24 weeks. Laboratory measurements, including hepatic and renal function, lipid profiles, nutritional status, and HbA1c, FBG, fasting insulin, and fasting C-peptide levels, were obtained at baseline and at 12 and 24 weeks. The Homeostatic Model Assessment for Insulin Resistance (HOMA-IR) was calculated using the following formula: HOMA-IR = (fasting insulin level × FBG level)/22.5 [28]. Visceral fat area was determined using InBody 720 Body Composition Analyzer (BioSpace Co., Ltd., Seoul, Korea).

To evaluate weight loss after metabolic surgery, %TWL, Δweight, BMI, and ΔBMI were calculated for all participants at every follow-up visit. The following formulas were used: %TWL = ([baseline weight—weight at each follow-up visit]/baseline weight) × 100%; Δweight = (weight at each follow-up visit)—(baseline weight); BMI = body weight (kg)/height^2 (m^2); and ΔBMI = (BMI at each follow-up visit)—(baseline BMI).[25]

Primary and Secondary Outcomes

The primary outcome of our study was the change in %TWL. The secondary outcomes included the proportions of patients achieving %TWL of ≥ 5%, ≥ 10%, and ≥ 15%; changes in weight and BMI; proportions of patients achieving the endpoint of healthy weight (18.5 kg/m^2 ≤ BMI < 24 kg/m^2), BMI < 28 kg/m^2, LDL-C levels < 2.6 mmol/l, HbA1c levels < 6%, and BP < 120/80 mmHg; rate of complication alleviation; and occurrence rates of adverse and severe adverse events.

Statistical Analyses

Using a two-sample *t*-test and based on preliminary experimental data, the difference in the mean percentage of weight change between the liraglutide and control groups was 4.8%; the group mean standard deviation (SD) was 5.4%, and statistical significance was set at the 5% level. With a sample size of 23 individuals per group, ensuring > 90% power and accounting for an estimated 20% dropout rate, at least 56 participants (28 per group) needed to be recruited. PASS 15.0 (NCSS, LLC., Kaysville, Utah, USA) was used to perform the sample size calculation.

Statistical analyses were performed using IBM SPSS Statistics for Windows version 25.0 (IBM Corp., Armonk, NY, USA). Categorical and continuous variables are presented as percentages and mean ± SD, respectively. Categorical

variables are described using absolute and relative frequencies. Continuous variables at baseline were compared using independent sample *t*-tests. Linear regression models adjusted for baseline values were used to analyze the differences in changes between the liraglutide and control groups at 24 weeks. Within-group differences before and after the intervention were assessed using *t*-tests. Weight, BMI, and %TWL were examined for differences using repeated-measures analysis of variance. Categorical variables were analyzed using Pearson's chi-square test and Fisher's exact test. A significance level of 0.05 was applied to all hypothesis testing.

RESULTS

A total of 61 patients who completed the 24-week follow-up were enrolled (liraglutide 1.8 mg daily, $n=27$; control, $n=34$) (Fig. 1). Baseline clinical characteristics were similar between the two groups, except for lower ferritin levels in the liraglutide group than in the control group (82.1 ± 13.6 µg/l vs. 180.1 ± 32.9 µg/l, $P=0.01$) (Table 1).

Weight Loss at 6 Months Postoperatively

Patients in the liraglutide and control groups showed similar preoperative BMI, weight loss, and %TWL at 6 months postoperatively (Supplementary Material, Table S1).

Primary Outcome

At week 24, the liraglutide group exhibited a greater reduction in %TWL than the control group (11.6% ± 1.1% vs. 4.9% ± 1.0%), with an adjusted mean difference of 6.6% (95% confidence interval [CI], 3.7–9.6%, $P<0.01$; Fig. 2A). A higher %TWL was observed at week 4 in the liraglutide group than in the control group, which increased over time even after the discontinuation of liraglutide after 12 weeks (Fig. 2A).

Secondary Outcomes

A higher number of patients lost ≥ 5% or ≥ 10% of their body weight in the liraglutide group than in the control group (92.6% vs. 55.9% and 55.6% vs. 17.6%, respectively; $P<0.05$) (Fig. 2B). Remarkable reductions in weight and BMI were observed in both groups. The liraglutide group

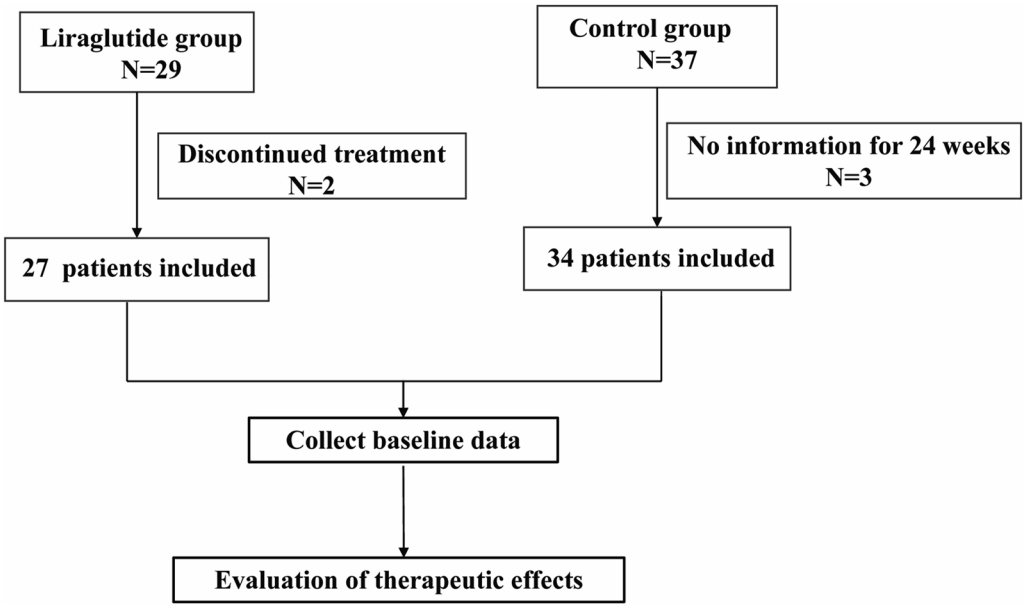


Fig. 1 Flow chart of the study

Table 1 Clinical characteristics of enrolled participants at baseline and 24 weeks

	Liraglutide group ($N = 27$)		Control group ($N = 34$)		*P*	Estimated mean treatment difference between two groups (95%CI)	*P*
	Baseline	24 weeks	Baseline	24 weeks	Baseline		24 weeks
Age (years)	30.0 ± 1.3	–	32.3 ± 1.4	–	0.24	–	–
Man (*n*, %)	5 (20.0)	–	14 (41.2)	–	0.10	–	–
SG (*n*, %)	23 (85.2)	–	30 (88.2)	–	1.00	–	–
RYGB (*n*, %)	4 (14.8)	–	4 (11.8)	–	1.00	–	–
Weight (kg)	92.9 ± 3.4	81.8 ± 2.8 *	92.6 ± 2.4	87.8 ± 2.2 *	0.95	− 6.2 (− 8.9 ~ − 3.4)	< 0.01
BMI (kg/m^2)	32.8 ± 1.2	28.2 ± 0.7 *	32.6 ± 0.7	31.0 ± 0.7 *	0.89	− 3.0 (− 4.2 ~ − 1.7)	< 0.01
VFA (cm^2)	168.2 ± 10.1	130.3 ± 7.4 *	157.9 ± 8.4	150.0 ± 8.8	0.44	− 25.8 (− 41.0 ~ − 10.5)	0.02
CAP (dB/m)	289.6 ± 6.1	268.4 ± 6.4 *	280.5 ± 5.0	279.3 ± 4.7	0.38	− 14.6 (− 30.1 ~ 0.9)	0.07
LSM (kPa)	7.1 ± 0.4	6.9 ± 0.3	7.7 ± 0.4	7.5 ± 0.5	0.25	− 0.5 (− 1.6 ~ 0.7)	0.45
SBP (mmHg)	124.2 ± 2.4	113.6 ± 2.2 *	128.6 ± 2.1	129.2 ± 3.1	0.18	− 14.1 (− 22.1 ~ − 6.1)	< 0.01
DBP (mmHg)	81.6 ± 1.5	75.9 ± 1.5 *	81.2 ± 1.9	82.5 ± 2.3	0.90	− 6.7 (− 12.6 ~ − 0.7)	0.03
HbA1c (%)	5.2 ± 0.1	5.0 ± 0.1 *	5.2 ± 0.1	5.2 ± 0.1	1.00	− 0.2(− 0.5 ~ 0.0)	0.06
FBG (mmol/l)	4.8 ± 0.0	4.5 ± 0.1 *	4.9 ± 0.1	4.7 ± 0.1 *	0.38	− 0.1 (− 0.3 ~ 0.1)	0.11
FINS (uIU/ml)	15.0 ± 1.6	9.7 ± 1.3 *	16.3 ± 1.5	11.4 ± 1.1 *	0.58	− 0.4 (− 3.2 ~ 2.3)	0.74
FCP (pmol/l)	1040.7 ± 74.1	860.2 ± 74.8 *	1041.2 ± 75.7	830.8 ± 48.4 *	1.00	59.9 (− 63.4 ~ 183.2)	0.33
HOMA-IR (mmol/l, uU/ml)	3.2 ± 0.4	2.0 ± 0.3 *	3.8 ± 0.5	2.4 ± 0.2 *	0.33	− 0.2 (− 1.0 ~ 0.6)	0.58
TG (mmol/l)	1.2 ± 0.1	1.0 ± 0.1 *	1.1 ± 0.1	0.9 ± 0.1	0.37	0.1 (− 0.1 ~ 0.3)	0.22
TC (mmol/l)	4.9 ± 0.2	4.6 ± 0.1 *	4.8 ± 0.2	4.8 ± 0.2	0.57	− 0.3 (− 0.7 ~ 0.0)	0.05
HDL-C(mmol/l)	1.3 ± 0.0	1.4 ± 0.0	1.4 ± 0.1	1.4 ± 0.1	0.66	0.0 (− 0.1 ~ 0.1)	0.05
LDL-C (mmol/l)	3.0 ± 0.1	2.7 ± 0.1 *	2.8 ± 0.2	2.9 ± 0.2	0.56	− 0.3 (− 0.6 ~ 0.0)	0.15
ALT (U/l)	21.4 ± 4.4	15.4 ± 1.8	27.1 ± 3.3	17.2 ± 1.7 *	0.29	− 1.4 (− 6.3 ~ 3.5)	0.57
AST (U/l)	18.4 ± 2.0	14.5 ± 0.5	22.8 ± 2.9	16.7 ± 0.7 *	0.23	− 2.0 (− 3.8 ~ 0.2)	0.03
GGT (U/l)	22.9 ± 2.4	18.5 ± 2.2 *	29.1 ± 6.0	21.2 ± 1.8 *	0.39	− 2.1 (− 7.6 ~ 3.3)	0.44

Table 1 continued

	Liraglutide group (*N* = 27)		Control group (*N* = 34)		*P*	Estimated mean treatment difference between two groups (95%CI)	*P*
	Baseline	24 weeks	Baseline	24 weeks	Baseline		24 weeks
UA (mmol/l)	377.3 ± 11.1	359.9 ± 11.0	363.5 ± 22.0	357.5 ± 12.2	0.61	− 0.8 (− 32.5 ~ 30.8)	1.00
Albumin (g/l)	44.3 ± 0.5	42.6 ± 0.6*	44.2 ± 0.4	42.8 ± 0.6	0.81	− 0.3 (− 1.7 ~ 1.1)	0.67
Ca (mmol/l)	2.5 ± 0.0	2.5 ± 0.0	2.5 ± 0.0	2.5 ± 0.0	0.64	0.0 (0.0– 0.1)	0.39
P (mmol/l)	1.4 ± 0.1	1.3 ± 0.0	1.2 ± 0.0	1.2 ± 0.0	0.13	0.0 (0.0– 0.1)	0.36
Ferritin (ug/l)	82.1 ± 13.6	82.7 ± 18.2	180.1 ± 32.9	139.9 ± 25.3	0.01	17.2 (− 16.0 ~ 50.5)	0.30
25-(OH)-D (ng/l)	20.9 ± 1.2	20.7 ± 0.9	21.0 ± 1.1	22.4 ± 1.5	0.94	− 1.5 (− 4.2 ~ 1.2)	0.27
VB12 (pg/ml)	572.2 ± 43.9	500.5 ± 28.6*	704.2 ± 50.7	697.3 ± 47.2	0.06	− 122.6 (− 219.0 ~ − 26.2)	0.02
Folate (pg/ml)	16.1 ± 1.2	15.4 ± 1.1	14.2 ± 1.2	13.1 ± 1.0	0.27	0.6 (− 3.1 ~ 4.2)	0.76
Obesity-related complications							
T2DM (*n*, %)	0 (0)	0 (0)	1 (2.9)	1 (2.9)	1.00	–	1.00
NAFLD (*n*, %)	25 (92.6)	8 (29.6)*	30 (88.2)	28 (82.4)	0.69	–	< 0.01
Hypertension (*n*, %)	6 (22.2)	1 (3.7)	16 (47.1)	14(41.2)	0.06	–	< 0.01
Dyslipidemia (*n*, %)	11 (44.0)	8 (29.6)	8 (32.0)	11 (32.4)	0.56	–	1.00
Hyperuricemia (*n*, %)	8 (29.6)	1 (3.7)*	10 (29.4)	4 (11.8)	1.00	–	0.37

ALT alanine transaminase, *AST* aspartate aminotransferase, *FBG* fasting blood glucose, *BMI* body weight index, *CAP* ultrasound attenuation parameter, *DBP* diastolic blood pressure, *FCP* fasting C-peptide, *FINS* fasting insulin, *GGT* gamma-glutamyltransferase, *HbA1c* glycated hemoglobin, *HDL-C* high-density lipoprotein cholesterol, *HOMA-IR* Homeostatic Model Assessment for Insulin Resistance, *LDL-C* low-density lipoprotein cholesterol, *LSM* liver stiffness measurement, *NAFLD* non-alcoholic fatty liver disease, *RYGB* Roux-en-Y gastric bypass, *SBP* systolic blood pressure, *SG* sleeve gastrectomy, *TG* triglyceride, *TC* total cholesterol, *T2DM* type 2 diabetes mellitus, *UA* uric acid, *VB12* vitamin 12, *VFA* visceral fat area, *$P < 0.05$, compared with baseline within the group

showed a substantial collective weight loss of 11.1 ± 1.2 kg, accompanied by a notable BMI decrease of 4.6 ± 0.8 kg/m^2. In comparison, the control group exhibited a weight reduction of 4.9 ± 1.1 kg, resulting in a BMI decrease of 1.8 ± 0.4 kg/m^2. The adjusted mean differences in the reduction of weight and BMI between the liraglutide and control groups were − 6.2 kg (95% CI − 8.9 to − 3.4, $P < 0.01$) and − 3.0 kg/m^2 (95% CI − 4.2 to − 1.7, $P < 0.01$), respectively (Table 1 and Fig. 3A, B). The downward trend for weight and BMI flattened at 12 weeks later in the control group but remained evident in the liraglutide group, even after drug discontinuation (Fig. 3A, B).

The liraglutide group had a significantly greater percentage of patients who reached the specific targets of BP < 120/80 mmHg (63.0%

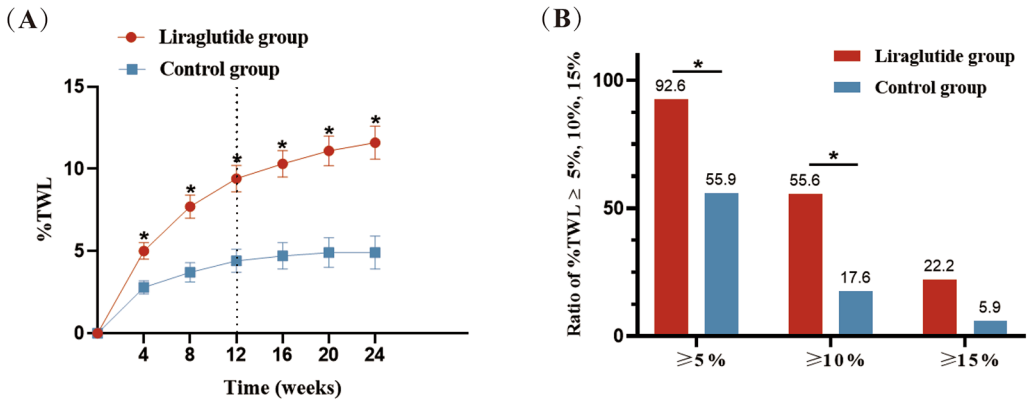


Fig. 2 Comparison of %TWL in the two groups. **A** Changes in %TWL at each follow-up visit. **B** Proportion of patients with ≥ 5%, 10%, and 15% weight loss at 24 weeks. TWL, total weight loss. The dashed line means liraglutide treatment was stopped

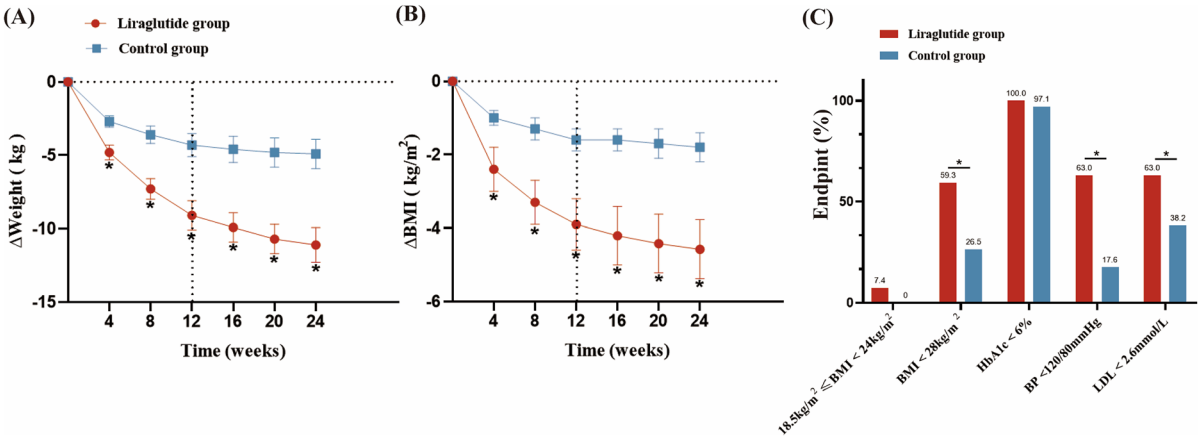


Fig. 3 Comparison of other indicators in two groups. **A** Changes in body weight at each follow-up visit. **B** Changes in BMI at each follow-up visit. **C** Proportion of patients achieving the composite endpoint of healthy weight, HbA1c < 6%, BP < 120/80 mmHg, and LDL-C levels < 2.6 mmol/l. *BMI* body mass index, *HbA1c* glycated hemoglobin, BP blood pressure, *LDL-C* low-density lipoprotein cholesterol. $^{*}P < 0.05$ compared with the control group. The dashed line means liraglutide treatment was stopped

vs. 17.6%) and LDL-C levels < 2.6 mmol/l (63.0% vs. 38.2%) than the control group (both $P < 0.05$; Fig. 3C). Patients in the liraglutide group achieved HbA1c levels < 6.0%, and 97.1% of patients in the control group achieved this target. Notably, the liraglutide group had a higher proportion of patients who achieved a BMI of < 28 kg/m^2 than did the control group (59.3% vs. 26.5%, $P < 0.05$). One patient in the liraglutide group achieved healthy weight, whereas none in the control group attained healthy weight (7.4% vs. 0.0%) (Fig. 3C).

Remission of Obesity-Related Comorbidities

From baseline to week 24, one patient in the control group was diagnosed with T2DM, which did not improve at 24 weeks later. No patient in the liraglutide group was diagnosed with T2DM. Compared with the control group, the

liraglutide group demonstrated notably elevated rates of remission for NAFLD (68.0% vs. 6.7%, $P<0.01$) and hypertension (83.3% vs. 12.5% $P<0.01$). Remarkably, compared with the control group, the liraglutide group exhibited an average reduction of – 14.1 mmHg (95% CI – 22.1 to – 6.1 mmHg, $P<0.01$) in SBP and – 6.7 mmHg (95% CI – 12.6 to – 0.7 mmHg, $P=0.03$) in DBP. Similar remission rates for dyslipidemia and hyperuricemia were found between the two groups (Table 1).

Nutritional Indicators

At baseline, ferritin levels were lower in the liraglutide group than in the control group; no noticeable disparities in the levels of albumin, ferritin, calcium, phosphorus, 25-(OH)-D, folate, and other crucial markers were observed between the two groups. The vitamin B12 values were lower in the liraglutide group than in the control group; however, they remained within the normal range (Table 1).

Adverse Events

The predominant adverse events observed in this study included nausea, hypotension, diarrhea, constipation, and cholelithiasis. Specifically, seven, one, two, and two participants in the liraglutide group experienced nausea, an episode of hypotension, diarrhea, and constipation, respectively. Furthermore, an ultrasound examination revealed cholelithiasis in one individual. Conversely, the control group had no discernible adverse reactions. Notably, severe adverse events were absent in both groups (Table 2).

DISCUSSION

Patients who remained obese 6 months after metabolic surgery had a lower possibility of achieving a lower BMI after 1 year because of increased food consumption, thereby limiting their potential of achieving greater metabolic benefits [5, 6]. The efficacy and safety of additional weight loss medication interventions at this point are a clinical challenge owing to a lack of clinical evidence. In this study, we showed that the administration of 1.8 mg of liraglutide for 12 weeks as an adjunct to a lifestyle intervention led to increased weight loss, improved cardiometabolic risk factors, and increased rates of remission for hypertension and NAFLD after 24 weeks. Furthermore, 92.6% and 55.6% of patients in the liraglutide group lost ≥ 5% and ≥ 10% of their baseline body weight, respectively, compared with 55.9% and 17.6% of patients in the control group. Minor adverse effects, primarily gastrointestinal, were reported with liraglutide, without any severe reactions and nutritional deficiencies. This evidence underscores the notion that timely liraglutide administration in individuals who remained obese 6 months after metabolic surgery can safely and efficaciously induce metabolic improvements while facilitating weight reduction.

To date, metabolic surgery remains the most effective therapy for morbid obesity; however, 25% of post-surgery patients experience inadequate weight loss outcomes after 2 years, and approximately 30% remain obese as time progresses [3, 4]. Historically, research on the timing of pharmaceutical interventions following

Table 2 Adverse events in the safety analysis of the enrolled population

	Liraglutide group	Control group	*P*
Nausea (*n*, %)	7 (25.9)	0 (0)	< 0.01
Hypotension (*n*, %)	1 (3.7)	0 (0)	0.44
Constipation (*n*, %)	2 (7.4)	0 (0)	0.19
Diarrhea (*n*, %)	2 (7.4)	0 (0)	0.19
Cholelithiasis (*n*, %)	1 (3.7)	0 (0)	0.44

metabolic surgery is limited, as most studies have implemented interventions > 1 year postoperatively when patients experience inadequate weight loss or weight regain [15, 20, 29, 30]. Initiating medication during a stable weight period rather than after noticeable weight regain may optimize postoperative weight loss outcomes [31]. A retrospective study showed that the lowest body weight was reached 1 year after metabolic surgery and that 80% of weight loss during the first postoperative year was observed within the initial 6 months [6, 16]. This trend may be associated with an increase in patients' dietary intake after 6 months postoperatively. A Swedish study reported that beginning from 6 months after surgery, the mean calorie intake of patients gradually increased from 1500 to 2000 kcal per day [32]. Therefore, we suggest that 6 months after surgery may be an ideal time for implementing pharmaceutical intervention. We found that initiating liraglutide 1.8 mg intervention 6 months postoperatively for those who remained obese helped achieve the lowest BMI (28.2 kg/m^2 vs. 31.0 kg/m^2) 1 year postoperatively. Although liraglutide was administered for only 12 weeks in our study, weight loss nadir was not reached for patients in this group at the end of the treatment period and the decline slope was better than that of the control group, thereby suggesting that earlier short-term adjuvant medication intervention is economical and effective for achieving lowest BMI.

Individuals with lower BMI after metabolic surgery have less possibility of weight regain and achieve greater and longer remission rates for obesity-related diseases [5]. In this study, the group administered liraglutide 1.8 mg exhibited a notable weight loss, elevated rate of NAFLD, and hypertension improvement.

Various weight loss medications, including phentermine, topiramate, orlistat, and GLP-1RAs, have been used to treat insufficient weight loss or weight regain in patients following metabolic surgery [33, 34]. Nonetheless, pertinent studies comparing the effects of different types of weight loss medications after metabolic surgery are currently limited. Some retrospective studies compared the weight loss effect of liraglutide and semaglutide in patients with weight regain after metabolic surgery; one study revealed that both liraglutide 3.0 mg daily and semaglutide 1.0 mg weekly resulted in effective weight loss in patients after metabolic surgery during a 1-year intervention; however, higher %TWL was achieved in patients receiving semaglutide 1.0 mg weekly than in those receiving liraglutide 3.0 mg daily (12.92% vs. 8.77%) [15]. Another study found that semaglutide 1.0 mg weekly or daily oral intake of semaglutide 14.0 mg resulted in greater %TWL than liraglutide 1.8 mg or 3.0 mg daily (9.8% vs. 7.3%) [30]. Liraglutide 1.8 mg was the highest dose approved in China at that time and was used in our study.

No nutritional deficiencies were noted in either group. The serum vitamin B12 levels decreased only in the liraglutide group at 24 weeks later and remained within the normal range, underscoring the importance of detecting nutritional status in patients after metabolic surgery, especially in those undergoing weight loss medication intervention.

Our study has several strengths. Patients who remained obese 6 months postoperatively were given weight loss drugs, which may help highlight earlier and more timely adjuvant weight loss medication intervention based on BMI rather than inadequate weight loss or weight regain with high BMI > 1 year postoperatively, thereby maximizing postoperative weight loss and resulting in lower body weight at the weight loss plateau for patients with severely obesity. The present study included a comprehensive assessment of cardiometabolic risk factors and obesity complications.

This study has some limitations. First, this retrospective study had a relatively small sample size and was conducted at a single center involving only the Chinese population. Second, 1.8 mg liraglutide was the highest dose approved in China and was used during our study, which is below the international maximum dose of 3.0 mg. Further limitations arise from the relatively short timeline of the intervention, which lasted for only 12 weeks.

CONCLUSION

In summary, 12 weeks of liraglutide 1.8 mg, combined with a lifestyle intervention, in

patients who remained obese 6 months following metabolic surgery was safe and well tolerated. This intervention resulted in effective reductions in body weight and improvement of cardiometabolic risk factors and obesity complications at 1 year postoperatively. This study provides valuable insights for maximizing benefits in patients with severe obesity postoperatively. In the future, studies on more effective weight loss medication intervention based on BMI within 1 year postoperatively might help in clarifying how to organically combine weight loss medication with metabolic surgery in patients with severe obesity to optimize weight loss after metabolic surgery.

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Author Contributions. Wenhuan Feng conceived the research and edited and revised the manuscript. Wenjuan Tang interpreted the data and edited and revised the manuscript. Yuanyuan Shen, Yuanhao Huang, and Yuqin Ouyang collected and analyzed the data and wrote the manuscript. Xuehui Chu performed metabolic surgery for the patients in the present study. Xinyue Xiang, Bingqing Zhang, and Tao Han collected the data. All authors approved the final submitted version of manuscript.

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Data Availability. Data generated and analyzed during this study are not publicly available but are available from the corresponding author upon reasonable request.

Declarations

Conflict of Interest. Yuanyuan Shen, Yuanhao Huang, and Yuqin Ouyang; Xinyue Xiang, Xuehui Chu, Bingqing Zhang, Tao Han, Wenjuan Tang, and Wenhuan Feng have no conflicts of interest to disclose.

Ethical Approval. The trial was conducted in accordance with the principles embodied in the Declaration of Helsinki and was approved by the Ethics Committee of the Affiliated Drum Tower Hospital, Medical School, Nanjing University (approval no.: 2017–030-08). All patient anonymity was preserved. All participants provided informed consent for their data potentially being used in the study.

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REFERENCES

1. Sjostrom L. Review of the key results from the Swedish Obese Subjects (SOS) trial—a prospective controlled intervention study of bariatric surgery. J Intern Med. 2013;273(3):219–34.

2. Pan XF, Wang L, Pan A. Epidemiology and determinants of obesity in China. Lancet Diabetes Endocrinol. 2021;9(6):373–92.

3. Brissman M, Beamish AJ, Olbers T, Marcus C. Prevalence of insufficient weight loss 5 years after Roux-en-Y gastric bypass: metabolic consequences and prediction estimates: a prospective registry study. BMJ Open. 2021;11(3):e046407.

4. Adams TD, Davidson LE, Litwin SE, Kim J, Kolotkin RL, Nanjee MN, et al. Weight and metabolic outcomes 12 years after gastric bypass. N Engl J Med. 2017;377(12):1143–55.

5. Courcoulas AP, King WC, Belle SH, Berk P, Flum DR, Garcia L, et al. Seven-year weight trajectories and health outcomes in the longitudinal assessment of bariatric surgery (LABS) study. JAMA Surg. 2018;153(5):427–34.

6. Shen Y, Zhang B, Hu X, Zhang N, Huang Y, Han T, et al. Metabolic surgery results in greater metabolic benefits in patients who achieve healthy weight. Surg Obes Relat Dis. 2024.

7. El Ansari W, Elhag W. Weight regain and insufficient weight loss after bariatric surgery: definitions, prevalence, mechanisms, predictors, prevention and management strategies, and knowledge gaps-a scoping review. Obes Surg. 2021;31(4):1755–66.

8. Melton GB, Steele KE, Schweitzer MA, Lidor AO, Magnuson TH. Suboptimal weight loss after gastric bypass surgery: correlation of demographics, comorbidities, and insurance status with outcomes. J Gastrointest Surg. 2008;12(2):250–5.

9. Cohen RV, Petry TB. How to address weight regain after bariatric surgery in an individualized way. Rev Endocr Metab Disord. 2023;24(5):993–1002.

10. Istfan NW, Lipartia M, Anderson WA, Hess DT, Apovian CM. Approach to the patient: management of the post-bariatric surgery patient with weight regain. J Clin Endocrinol Metab. 2021;106(1):251–63.

11. Yimcharoen P, Heneghan HM, Singh M, Brethauer S, Schauer P, Rogula T, et al. Endoscopic findings and outcomes of revisional procedures for patients with weight recidivism after gastric bypass. Surg Endosc. 2011;25(10):3345–52.

12. Saliba C, El Rayes J, Diab S, Nicolas G, Wakim R. Weight regain after sleeve gastrectomy: a look at the benefits of re-sleeve. Cureus. 2018;10(10):e3450.

13. Hansen D, Decroix L, Devos Y, Nocca D, Cornelissen V, Dillemans B, et al. Towards optimized care after bariatric surgery by physical activity and exercise intervention: a review. Obes Surg. 2020;30(3):1118–25.

14. Schwartz J, Chaudhry UI, Suzo A, Durkin N, Wehr AM, Foreman KS, et al. Pharmacotherapy in conjunction with a diet and exercise program for the treatment of weight recidivism or weight loss plateau post-bariatric surgery: a retrospective review. Obes Surg. 2016;26(2):452–8.

15. Murvelashvili N, Xie L, Schellinger JN, Mathew MS, Marroquin EM, Lingvay I, et al. Effectiveness of semaglutide versus liraglutide for treating post-metabolic and bariatric surgery weight recurrence. Obesity (Silver Spring). 2023;31(5):1280–9.

16. Torrego-Ellacuria M, Barabash A, Larrad-Sainz A, Hernandez-Nunez GM, Matia-Martin P, Perez-Ferre N, et al. Weight regain outcomes after bariatric surgery in the long-term follow-up: role of preoperative factors. Obes Surg. 2021;31(9):3947–55.

17. Alvarez V, Carrasco F, Cuevas A, Valenzuela B, Munoz G, Ghiardo D, et al. Mechanisms of long-term weight regain in patients undergoing sleeve gastrectomy. Nutrition. 2016;32(3):303–8.

18. Courcoulas AP, Christian NJ, Belle SH, Berk PD, Flum DR, Garcia L, et al. Weight change and health outcomes at 3 years after bariatric surgery among individuals with severe obesity. JAMA. 2013;310(22):2416–25.

19. van Can J, Sloth B, Jensen CB, Flint A, Blaak EE, Saris WH. Effects of the once-daily GLP-1 analog liraglutide on gastric emptying, glycemic parameters, appetite and energy metabolism in obese, non-diabetic adults. Int J Obes (Lond). 2014;38(6):784–93.

20. Mok J, Adeleke MO, Brown A, Magee CG, Firman C, Makahamadze C, et al. Safety and efficacy of liraglutide, 3.0 mg, once daily vs placebo in patients with poor weight loss following metabolic surgery: the BARI-OPTIMISE randomized clinical trial. JAMA Surg. 2023;158(10):1003–11.

21. Vinciguerra F, Piazza L, Di Stefano C, Degano C, Pulvirenti A, Baratta R, et al. High-dose liraglutide improves metabolic syndrome in poor responders to bariatric surgery. Front Nutr. 2023;10:1183899.

22. Bull FC, Al-Ansari SS, Biddle S, Borodulin K, Buman MP, Cardon G, et al. World Health Organization

2020 guidelines on physical activity and sedentary behaviour. Br J Sports Med. 2020;54(24):1451–62.

23. Society CD. Guideline for the prevention and treatment of type 2 diabetes mellitus in China (2020 edition). Chin J Diabetes. 2021;41(5):482–548.

24. Brethauer SA, Kim J, el Chaar M, Papasavas P, Eisenberg D, Rogers A, et al. Standardized outcomes reporting in metabolic and bariatric surgery. Surg Obes Relat Dis. 2015;11(3):489–506.

25. Brethauer SA, Kim J, El Chaar M, Papasavas P, Eisenberg D, Rogers A, et al. Standardized outcomes reporting in metabolic and bariatric surgery. Obes Surg. 2015;25(4):587–606.

26. Qu Y, Song YY, Chen CW, Fu QC, Shi JP, Xu Y, et al. Diagnostic performance of FibroTouch ultrasound attenuation parameter and liver stiffness measurement in assessing hepatic steatosis and fibrosis in patients with nonalcoholic fatty liver disease. Clin Transl Gastroenterol. 2021;12(4):e00323.

27. Wu J, Qiu L, Cheng XQ, Xu T, Wu W, Zeng XJ, et al. Hyperuricemia and clustering of cardiovascular risk factors in the Chinese adult population. Sci Rep. 2017;7(1):5456.

28. Matthews DR, Hosker JP, Rudenski AS, Naylor BA, Treacher DF, Turner RC. Homeostasis model assessment: insulin resistance and beta-cell function from fasting plasma glucose and insulin concentrations in man. Diabetologia. 1985;28(7):412–9.

29. Wharton S, Kuk JL, Luszczynski M, Kamran E, Christensen RAG. Liraglutide 30 mg for the management of insufficient weight loss or excessive weight regain post-bariatric surgery. Clin Obes. 2019;9(4):12323.

30. Jensen AB, Renström F, Aczél S, Folie P, Biraima-Steinemann M, Beuschlein F, et al. Efficacy of the glucagon-like peptide-1 receptor agonists liraglutide and semaglutide for the treatment of weight regain after bariatric surgery: a retrospective observational study. Obes Surg. 2023;33(4):1017–25.

31. Stanford FC, Alfaris N, Gomez G, Ricks ET, Shukla AP, Corey KE, et al. The utility of weight loss medications after bariatric surgery for weight regain or inadequate weight loss: a multi-center study. Surg Obes Relat Dis. 2017;13(3):491–500.

32. Clapp B, Wynn M, Martyn C, Foster C, O'Dell M, Tyroch A. Long term (7 or more years) outcomes of the sleeve gastrectomy: a meta-analysis. Surg Obes Relat Dis. 2018;14(6):741–7.

33. Redmond IP, Shukla AP, Aronne LJ. Use of weight loss medications in patients after bariatric surgery. Curr Obes Rep. 2021;10(2):81–9.

34. Rubino DM, Greenway FL, Khalid U, O'Neil PM, Rosenstock J, Sørrig R, et al. Effect of weekly subcutaneous semaglutide vs daily liraglutide on body weight in adults with overweight or obesity without diabetes: the STEP 8 randomized clinical trial. JAMA. 2022;327(2):138–50.